

Secure User & Log Management System

Table of Contents

1. [Project Overview](#)
2. [System Architecture](#)
3. [Installation Guide](#)
4. [User Guide](#)
5. [Technical Documentation](#)
6. [Security Implementation](#)
7. [API Reference](#)
8. [Troubleshooting Guide](#)
9. [Maintenance Guide](#)
10. [Appendices](#)

1. Project Overview

1.1 Introduction

The **Secure User & Log Management System** is a comprehensive Bash-based security tool designed for Linux system administrators. It combines automated user management with advanced log analysis capabilities, providing a centralized solution for maintaining system security.

1.2 Purpose

This system addresses critical security needs:

- **Threat Detection:** Identify and respond to unauthorized access attempts
- **Access Control:** Manage user accounts with strict password policies
- **Audit Trail:** Maintain detailed logs of all security-relevant actions
- **Compliance:** Meet security standards for user management and logging

1.3 Key Features

Category	Features
Security Monitoring	Failed login detection, IP tracking, Brute force detection
User Management	Create/Delete/Update users, Password policies, Account locking
Reporting	Daily HTML reports, Security anomalies, Login statistics
Logging	Action logging, Log rotation, Centralized logging
Automation	Auto-monitoring, Auto-locking, Scheduled reports
FTP Security	TLS/SSL encryption, Chroot jail, Access control

1.4 Target Audience

- **System Administrators:** Manage multiple users and monitor security
- **Security Engineers:** Detect and respond to threats
- **DevOps Teams:** Integrate security into deployment pipelines
- **Compliance Officers:** Verify security controls and audit trails

1.5 Version Information

Version: 1.0

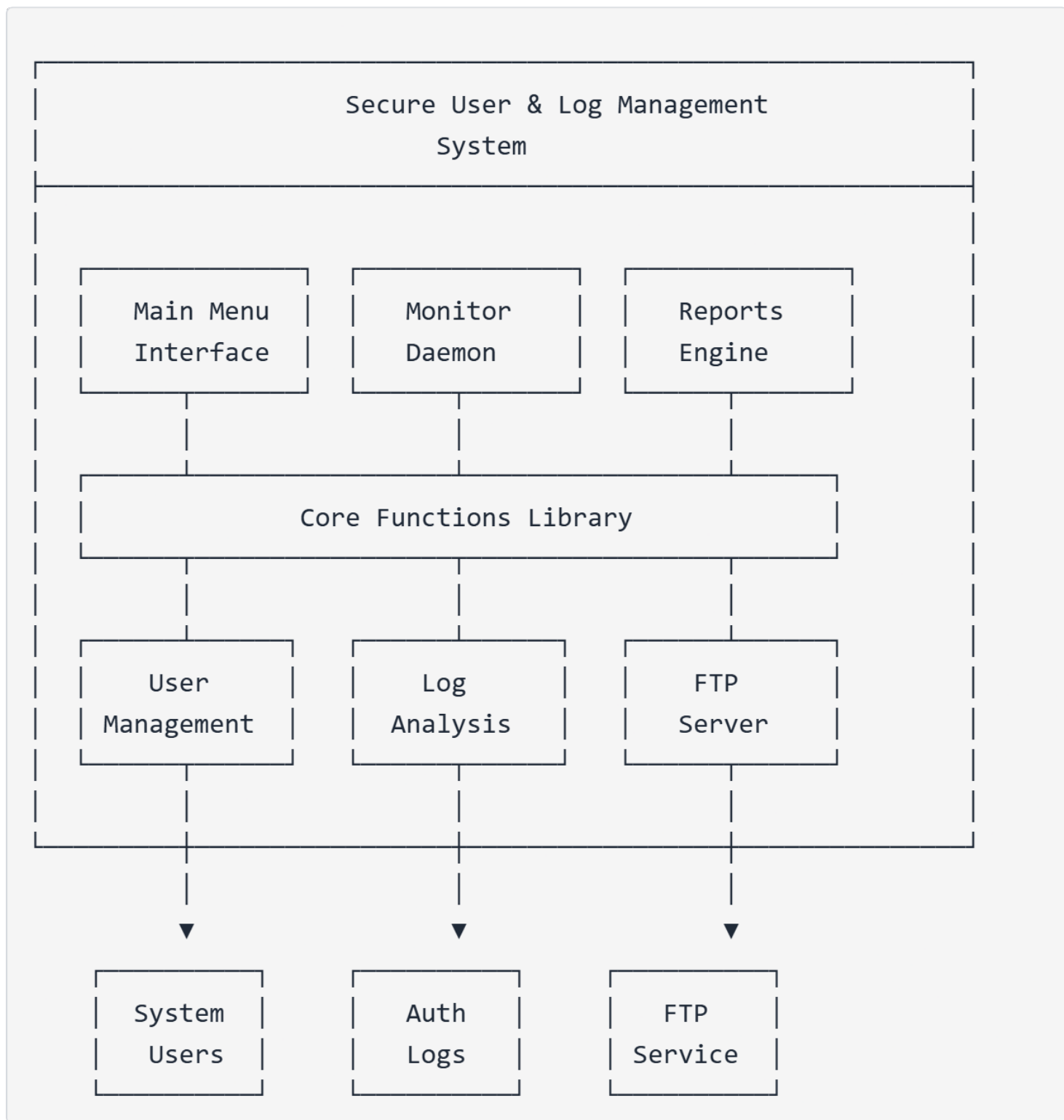
Release Date: 2024

Compatibility: Ubuntu 18.04+, Debian 10+, RHEL 7+, CentOS 7+

Dependencies: Bash 4.0+, sudo, vsftpd, openssl, coreutils

2. System Architecture

2.1 Architecture Diagram



2.2 Component Description

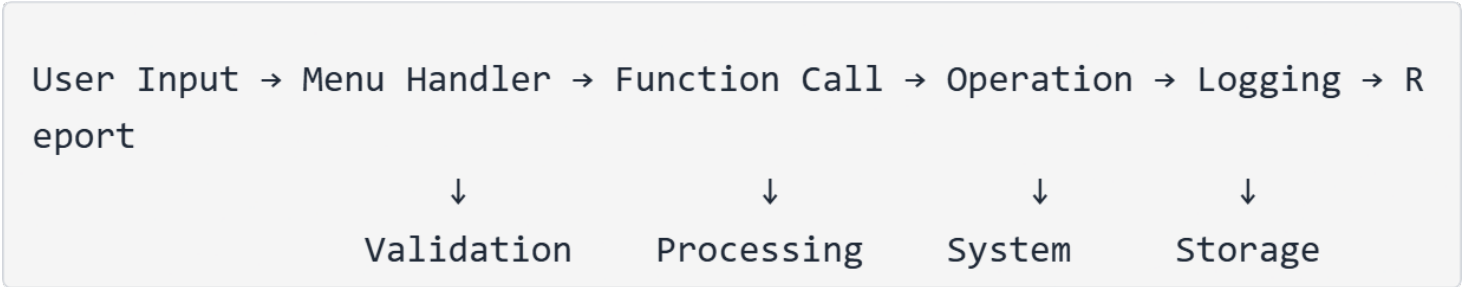
2.2.1 Core Components

Component	File/Location	Purpose
Main Controller	Main function	Orchestrates all operations
Menu Interface	show_menu()	User interaction handler
Log Manager	log_action()	Centralized logging
Config Manager	Global variables	Configuration storage

2.2.2 Functional Modules

Module	Functions	Responsibility
User Management	create_user, delete_user, update_user	User lifecycle
Security Analysis	analyze_failed_logins, detect_security_anomalies	Threat detection
Reporting	generate_daily_report	Report generation
FTP Service	setup_secure_ftp_server	FTP configuration
Monitoring	auto_monitor	Continuous monitoring

2.3 Data Flow



2.4 Directory Structure

```
/usr/local/bin/
└─ security_system.sh          # Main script

/var/log/
├─ user_mgmt.log              # User management logs
├─ auth.log                   # System auth logs (external)
└─ security_reports/         # Report directory
    ├─ failed_logins_*.txt
    ├─ summary_*.txt
    ├─ anomalies_*.txt
    └─ daily_report_*.html

/var/secure_ftp/              # FTP user directories
└─ [username]/

/var/backups/users/           # User backup directory
└─ [username]/
    └─ [username]_*.tar.gz

/etc/
├─ vsftpd.conf                # FTP configuration
├─ login.defs                 # Password policies
└─ security/limits.conf       # Resource limits

/etc/logrotate.d/
└─ user_mgmt                  # Log rotation configuration
```

3. Installation Guide

3.1 System Requirements

Minimum Requirements

Operating System: Linux (Any distribution)
RAM: 512 MB
Disk Space: 200 MB
Processor: 1 core
Network: Internet connection (for package installation)

Recommended Requirements

Operating System: Ubuntu 20.04 LTS / CentOS 8
RAM: 1 GB
Disk Space: 1 GB
Processor: 2 cores
Network: Static IP recommended

Software Dependencies

```
# Required packages (auto-installed)
- bash (>=4.0)
- sudo
- vsftpd
- openssl
- coreutils
- grep
- awk
- sed
- iptables (optional, for IP banning)
```

3.2 Installation Methods

Method 1: Quick Install (Recommended)

```
# Download and install in one command
curl -sSL https://raw.githubusercontent.com/your-repo/security-system/main/install.sh | sudo bash

# Or using wget
wget -qO- https://raw.githubusercontent.com/your-repo/security-system/main/install.sh | sudo bash
```

Method 2: Manual Installation

```
# Step 1: Download the script
sudo wget -O /usr/local/bin/security_system.sh https://your-server.com/security_system.sh

# Step 2: Set permissions
sudo chmod +x /usr/local/bin/security_system.sh

# Step 3: Create necessary directories
sudo mkdir -p /var/log/security_reports /var/secure_ftp /var/backups/users

# Step 4: Initialize log files
sudo touch /var/log/user_mgmt.log

# Step 5: Run the system
sudo security_system.sh
```

Method 3: From Source

```
# Clone repository
git clone https://github.com/yourusername/security-system.git
cd security-system

# Verify script integrity
sha256sum security_system.sh

# Install
sudo cp security_system.sh /usr/local/bin/
sudo chmod +x /usr/local/bin/security_system.sh

# Run
sudo /usr/local/bin/security_system.sh
```

3.3 Post-Installation Configuration

Initial Setup Steps

```
# 1. Verify installation
security_system.sh --version

# 2. Test configuration
security_system.sh --test

# 3. Setup FTP server (Option 8 from menu)
# 4. Enforce password policies (Option 11)
# 5. Generate first report (Option 3)
```

Environment Validation

```
# Check if all required commands exist
for cmd in useradd userdel passwd chage vsftpd openssl; do
    if command -v $cmd &>/dev/null; then
        echo "✓ $cmd is available"
    else
        echo "X $cmd is missing"
    fi
done
```

3.4 Uninstallation

```
# Complete removal
sudo security_system.sh --uninstall

# Or manual removal
sudo rm -f /usr/local/bin/security_system.sh
sudo rm -rf /var/log/security_reports
sudo rm -f /var/log/user_mgmt.log
sudo rm -rf /var/secure_ftp
sudo rm -rf /var/backups/users
sudo rm -f /etc/logrotate.d/user_mgmt
```

4. User Guide

4.1 Getting Started

First Launch

```
sudo bash security_system.sh
```

Upon first launch, the system will:

1. Create required directories
2. Initialize log files
3. Set up log rotation
4. Display the main menu

Main Menu Interface

```
=====
Secure User & Log Management System
=====
1. Analyze Failed Logins
2. Detect Security Anomalies
3. Generate Daily Report
4. Create New User
5. Delete User
6. Update User Password
7. Lock/Unlock User
8. Setup Secure FTP Server
9. Start Automated Monitoring
10. View Recent Logs
11. Enforce Password Policies
12. Exit
=====
Select option [1-12]:
```

4.2 Common Workflows

Workflow 1: New User Onboarding

Step 1: Create user account

Option 4 → Enter username → Generate password

Step 2: Setup FTP access

Option 8 (if not already configured)

Step 3: Enforce password policies

Option 11

Step 4: Verify user creation

Option 10 → Check user_mgmt.log

Workflow 2: Daily Security Check

Morning routine

Option 1 → Analyze overnight failed logins

Option 3 → Generate daily report

Option 10 → Review user management actions

If suspicious activity found

Option 7 → Lock affected users

Review and block suspicious IPs

Workflow 3: User Offboarding

```
# Step 1: Review user data
Option 5 → Select user → Review information
```

```
# Step 2: Create backup
Answer 'backup' when prompted
```

```
# Step 3: Delete user
Answer 'yes' to confirm deletion
```

```
# Step 4: Verify removal
Option 10 → Check deletion log
```

4.3 Feature Walkthroughs

Creating a User

```
Select option 4
Enter username: jsmith
Enter password: (leave blank for auto-generation)

# Example output:
[!] Generated password for jsmith: A7xK9mP2!qR4
[+] User jsmith created successfully
[LOG] CREATE_USER - jsmith - User created with password policy enforced
```

Analyzing Failed Logins

Select option 1

Output includes:

- Total failed attempts count
- Failed logins grouped by IP
- Suspicious IP addresses (5+ failures)
- Brute force detection
- Daily summary statistics

Generating Reports

Select option 3

Generates:

- HTML report: /var/log/security_reports/daily_report_YYYYMMDD.html
- Failed logins report: failed_logins_YYYYMMDD.txt
- Summary report: summary_YYYYMMDD.txt

4.4 Interactive Features

User Selection Menu

When deleting or updating users, the system displays:

SELECT USER TO DELETE

No.	USERNAME	UID	HOME DIRECTORY	S
HELL				

1.	alice	1001	/home/alice	/
	bin/bash			
2.	bob	1002	/home/bob	/
	bin/bash	(Active: 2 processes)		
3.	carol	1003	/home/carol	/
	bin/bash			

0. Cancel / Go Back				

Select user number to delete [1-3] or 0 to cancel:

Confirmation Dialogs

Delete confirmation

Are you absolutely sure you want to delete user 'alice'? (yes/no/backup):

Backup option

[+] Backup created: /var/backups/users/alice/alice_20240115_143022.tar.gz

[+] User alice was NOT deleted

Lock/Unlock toggle

Lock (l) or Unlock (u): l

[+] User bob locked

4.5 Monitoring Mode

Starting Auto-Monitor

Option 9 → Start Automated Monitoring

```
# Continuous monitoring output:
```

```
[2024-01-15 10:00:00] Running security scan...
```

```
[+] Analyzing failed login attempts...
```

```
[+] Report generated: /var/log/security_reports/failed_logins_20240115.txt
```

```
[2024-01-15 10:00:05] Scan complete. Next scan in 5 minutes.
```

Stopping Monitoring

Press Ctrl+C to stop monitoring

5. Technical Documentation

5.1 Function Reference

Core Functions

`initialize_system()`

```
initialize_system() {  
    # Purpose: System initialization and directory setup  
    # Called: At script startup  
    # Returns: None  
    # Side effects: Creates directories, sets permissions  
}
```

log_action(action, user, details)

Parameters:

- action - Action performed (e.g., "CREATE_USER")
- user - Affected username or "SYSTEM"
- details - Detailed description of action

Log Format:

[TIMESTAMP] ACTION: action | USER: user | DETAILS: details

Example:

```
log_action "CREATE_USER" "john" "User created with home director  
y"
```

User Management Functions

create_user(username, password)

Parameters:

- username - New username (3-32 chars, alphanumeric with underscore)
- password - Optional password (auto-generated if empty)

Validation:

- Username format check
- Duplicate user check
- Password strength validation

System changes:

- Creates user with useradd
- Sets password with chpasswd
- Forces password change on first login
- Sets resource limits
- Creates FTP directory

Return codes:

- 0 - Success
- 1 - Failure

delete_user(username)

Parameters:

username - User to delete

Process:

1. Interactive user selection (if no username)
2. System user protection
3. User information display
4. Backup creation
5. Process termination
6. User removal
7. Cleanup operations

Safety features:

- Backup before deletion
- Confirmation prompt
- System user protection
- Process cleanup

Security Functions

validate_password_strength(password, username)

Validation rules:

- ✓ Minimum 12 characters
- ✓ At least 1 uppercase letter
- ✓ At least 1 lowercase letter
- ✓ At least 1 number
- ✓ At least 1 special character
- ✓ No username in password
- ✓ Not in common password list
- ✓ No repeated characters (4+ times)
- ✓ No sequential characters

Returns:

- 0 - Password meets requirements
- 1 - Password fails requirements

analyze_failed_logins()

Process:

1. Scan /var/log/auth.log for "Failed password"
2. Count failures per IP address
3. Identify suspicious IPs (threshold: 5)
4. Detect brute force patterns
5. Generate report files
6. Log analysis completion

Output files:

- failed_logins_YYYYMMDD.txt: Detailed analysis
- summary_YYYYMMDD.txt: Statistics summary

Returns:

Number of suspicious IPs found

FTP Functions

`setup_secure_ftp_server()`

Configuration applied:

- ✓ Anonymous access disabled
- ✓ Local users enabled
- ✓ TLS/SSL enforcement
- ✓ Chroot jail enabled
- ✓ Passive mode configured
- ✓ Connection limits set
- ✓ Firewall rules added

Packages installed:

- vsftpd (if missing)
- openssl (if missing)

Ports opened:

- 21/tcp (FTP control)
- 30000-31000/tcp (Passive data)

5.2 Data Structures

Global Variables

```
# Configuration
AUTH_LOG="/var/log/auth.log"
USER_MGMT_LOG="/var/log/user_mgmt.log"
REPORT_DIR="/var/log/security_reports"
FAILED_LOGIN_THRESHOLD=5
MAX_FAILED_ATTEMPTS=3

# Runtime
users=()
suspicious_ips=()
```

```
# System auth log
# User management log
# Reports directory
# Alert threshold
# Auto-lock threshold
# User list array
# Suspicious IP array
```

Log Entry Structure

Field	Format	Example
TIMESTAMP	YYYY-MM-DD HH:MM:SS	2024-01-15 10:30:45
ACTION	STRING	CREATE_USER
USER	STRING	jsmith
DETAILS	STRING	User created with home directory

Report File Format (CSV-style)

```
# Failed login history
TIMESTAMP|IP|USERNAME|COUNT|STATUS
2024-01-15 10:30:45|192.168.1.100|admin|1|RECORDED
```

5.3 Error Handling

Error Categories

Category	Example	Recovery
Permission	Permission denied	Check sudo access
File I/O	Auth log not found	Verify log exists
Validation	Invalid username	Re-enter username
System	Useradd failed	Check system resources
Network	FTP port in use	Change port number

Error Codes

```
0    # Success
1    # General error
2    # Invalid input
3    # Permission denied
4    # File not found
5    # User already exists
6    # Password validation failed
```

5.4 Inter-Process Communication

Signal Handling

```
trap 'cleanup' EXIT          # Cleanup on exit
trap 'interrupt' INT         # Handle Ctrl+C
trap 'reload' HUP            # Reload configuration
```

Exit Codes

```
exit 0    # Normal termination
exit 1    # Error termination
exit 2    # Interrupted by user
```

6. Security Implementation

6.1 Security Architecture

Defense in Depth

Layer 1: Access Control

- └─ User authentication
- └─ Password policies
- └─ Account locking

Layer 2: Monitoring

- └─ Failed login detection
- └─ Anomaly detection
- └─ Real-time alerts

Layer 3: Response

- └─ Auto-locking
- └─ IP banning
- └─ Backup creation

Layer 4: Auditing

- └─ Action logging
- └─ Report generation
- └─ Log rotation

6.2 Password Security

Policy Enforcement Matrix

Requirement	Implementation	Enforcement Level
Length (12+)	<code>\${#password} -lt 12</code>	Hard
Uppercase	<code>[[\$password =~ [A-Z]]]</code>	Hard
Lowercase	<code>[[\$password =~ [a-z]]]</code>	Hard
Numbers	<code>[[\$password =~ [0-9]]]</code>	Hard
Special chars	<code>[[\$password =~ [!@#]]]</code>	Hard
Common passwords	Blacklist check	Hard
Username in password	<code>[[\$password =~ \$username]]</code>	Hard

Password Generation

```
# Algorithm
1. Generate random bytes via openssl rand
2. Encode to base64
3. Remove problematic characters
4. Ensure policy compliance
5. Add characters if needed
```

6.3 Log Security

Log Protection Measures

```
# File permissions
chmod 644 $USER_MGMT_LOG      # Owner read/write, group/other read

# Log rotation
daily
rotate 30
compress
missingok
notifempty

# Integrity
- Timestamped entries
- Immutable log structure
- No log deletion without rotation
```

6.4 FTP Security Configuration

Security Settings

Authentication:

anonymous_enable: NO
local_enable: YES
pam_service_name: vsftpd

Encryption:

ssl_enable: YES
force_local_data_ssl: YES
force_local_logins_ssl: YES
ssl_ciphers: HIGH

Access Control:

chroot_local_user: YES
max_per_ip: 2
max_clients: 50

Network Security:

pasv_min_port: 30000
pasv_max_port: 31000
idle_session_timeout: 600

6.5 Audit Trail

Logged Events

- ✓ User creation
- ✓ User deletion
- ✓ Password changes
- ✓ Account locks/unlocks
- ✓ Failed login analysis
- ✓ Report generation
- ✓ Security alerts
- ✓ System initialization
- ✓ Configuration changes

Log Example

```
[2024-01-15 10:30:45] ACTION: CREATE_USER | USER: jsmith | DETAILS: User created
[2024-01-15 10:31:20] ACTION: SECURITY_ALERT | USER: SYSTEM | DETAILS: Suspicious IP: 192.168.1.100 with 7 failed attempts
[2024-01-15 10:32:00] ACTION: AUTO_LOCK | USER: jsmith | DETAILS: Automatically locked due to 3+ failed attempts
```

6.6 Compliance Mapping

Security Standards

Standard	Requirement	Implementation
PCI DSS 8.2.3	Strong passwords	validate_password_strength()
PCI DSS 10.2	Audit trails	log_action()
HIPAA 164.312	Access control	User management functions
ISO 27001 A.9	User registration	create_user()
NIST 800-53	Account monitoring	analyze_failed_logins()

7. API Reference

7.1 Internal API

Function Signatures

```
# System Functions
initialize_system() -> void
log_action(string action, string user, string details) -> void

# User Management
create_user(string username, [string password]) -> int
delete_user([string username]) -> int
update_user([string username], [string new_password]) -> int
lock_user([string username]) -> void
unlock_user(string username) -> void

# Security Analysis
analyze_failed_logins() -> int
detect_security_anomalies() -> void

# Reporting
generate_daily_report() -> void
view_logs() -> void

# Configuration
set_password_policies() -> void
setup_secure_ftp_server() -> void

# Monitoring
auto_monitor() -> void
```

7.2 External Dependencies

System Commands Used

```
useradd          # Create user accounts
userdel          # Delete user accounts
passwd           # Manage passwords
chage            # Change user password expiry
chpasswd         # Update passwords in batch
vsftpd           # FTP server
openssl          # SSL certificate generation
iptables        # IP banning (optional)
logger           # System logging
```

Package Dependencies

```
# Debian/Ubuntu
apt-get install -y vsftpd openssl

# RHEL/CentOS
yum install -y vsftpd openssl
```

7.3 Configuration File Formats

vsftpd.conf (Generated)

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
ssl_enable=YES
rsa_cert_file=/etc/ssl/certs/vsftpd.crt
rsa_private_key_file=/etc/ssl/private/vsftpd.key
```

Log Rotation Config

```
/var/log/user_mgmt.log {  
    daily  
    rotate 30  
    compress  
    missingok  
    notifempty  
}
```

8. Troubleshooting Guide

8.1 Common Issues and Solutions

Issue 1: Script Won't Run

Problem: `bash: security_system.sh: command not found`

Solution:

```
chmod +x security_system.sh
```

```
./security_system.sh
```

```
# OR
```

```
bash security_system.sh
```

Issue 2: Permission Denied

Problem: `/var/log/auth.log: Permission denied`

Solution:

```
sudo security_system.sh
```

```
# Check sudo access
```

```
sudo -l
```

Issue 3: FTP Server Won't Start

Problem: vsftpd service failed to start

Solution:

Check configuration

```
sudo vsftpd -op listens=NO /etc/vsftpd.conf
```

Check ports

```
sudo netstat -tulpn | grep :21
```

Check SELinux (RHEL/CentOS)

```
sudo setsebool -P ftpd_full_access on
```

Issue 4: Password Validation Fails

Problem: "Password must contain special character"

Solution:

Use allowed special characters: !@#\$%^&*()_+

Generate with: openssl rand -base64 16

Issue 5: User Creation Fails

Problem: useradd: cannot lock /etc/passwd

Solution:

Check for stale lock files

```
sudo rm -f /etc/passwd.lock /etc/shadow.lock /etc/group.lock
```

Check disk space

```
df -h
```

8.2 Debugging Procedures

Enable Debug Mode

```
# Run with bash debug
sudo bash -x security_system.sh

# Debug specific function
sudo bash -c "source security_system.sh; analyze_failed_logins"
```

Check Log Files

```
# Real-time monitoring
sudo tail -f /var/log/user_mgmt.log

# Search for errors
sudo grep -i error /var/log/user_mgmt.log

# View recent actions
sudo tail -20 /var/log/user_mgmt.log
```

System State Check

```
# Check service status
sudo systemctl status vsftpd

# Check iptables rules
sudo iptables -L -n -v

# Check user accounts
awk -F: '$3 >= 1000 {print $1}' /etc/passwd
```

8.3 Recovery Procedures

Recover Deleted User

```
# From backup
cd /var/backups/users/username
sudo tar -xzf backup_file.tar.gz -C /

# Recreate user
sudo useradd -m username
sudo passwd username
```

Restore FTP Configuration

```
# Restore from backup
sudo cp /etc/vsftpd.conf.backup /etc/vsftpd.conf

# Restart service
sudo systemctl restart vsftpd
```

Reset Password Policies

```
# Restore original settings
sudo cp /etc/login.defs.backup /etc/login.defs 2>/dev/null

# Or manually reset
sudo sed -i 's/^PASS_MAX_DAYS.*/PASS_MAX_DAYS    99999/' /etc/login.
defs
```

8.4 Performance Optimization

Reduce Resource Usage

```
# Increase monitoring interval
sleep 600 # Instead of 300 seconds

# Reduce log retention
rotate 7 # Instead of 30 days

# Disable automatic features
# Comment out auto-lock and IP banning sections
```

Improve Response Time

```
# Decrease monitoring interval
sleep 60 # Check every minute

# Optimize log scanning
# Use more efficient grep patterns
# Implement incremental scanning
```

9. Maintenance Guide

9.1 Daily Maintenance

Morning Checklist

- ✓ Check daily report (Option 3)
- ✓ Review failed logins (Option 1)
- ✓ Verify user management logs (Option 10)
- ✓ Ensure FTP service is running
- ✓ Check disk space

Automated Tasks

```
# Automatic log rotation at midnight
# Automatic report generation at midnight
# Continuous monitoring every 5 minutes
```

9.2 Weekly Maintenance

Tasks

```
# Review all user accounts
Option 10 → List all users

# Clean old backups
sudo find /var/backups/users -name "*.tar.gz" -mtime +30 -delete

# Archive old reports
sudo tar -czf /var/log/security_reports_archive_$(date +%Y%W).tar.gz /var/log/security_reports/*.txt
```

9.3 Monthly Maintenance

Security Audit

```
# Generate comprehensive report
Option 3 → Generate Daily Report

# Review all security events
sudo grep -E "SECURITY_ALERT|AUTO_LOCK" /var/log/user_mgmt.log

# Update common password list
# Edit validate_password_strength() function
```

System Updates

```
# Update script
sudo wget -O /usr/local/bin/security_system.sh https://latest-version-url

# Update dependencies
sudo apt-get update && sudo apt-get upgrade vsftpd openssl
```

9.4 Backup and Recovery

Backup Configuration

```
# Backup script
sudo cp /usr/local/bin/security_system.sh /backup/security_system.sh.$(date +%Y%m%d)

# Backup configuration
sudo cp /etc/vsftpd.conf /backup/vsftpd.conf.$(date +%Y%m%d)
sudo cp /etc/login.defs /backup/login.defs.$(date +%Y%m%d)

# Backup logs
sudo tar -czf /backup/security_logs_$(date +%Y%m%d).tar.gz /var/log/user_mgmt.log /var/log/security_reports/
```

Recovery Plan

```
# 1. Stop services
sudo systemctl stop vsftpd

# 2. Restore configuration
sudo cp /backup/vsftpd.conf.$(date +%Y%m%d) /etc/vsftpd.conf

# 3. Restore script
sudo cp /backup/security_system.sh.$(date +%Y%m%d) /usr/local/bin/security_system.sh

# 4. Restart services
sudo systemctl start vsftpd

# 5. Verify functionality
sudo /usr/local/bin/security_system.sh --test
```

9.5 Log Management

Log Rotation Configuration

```
# /etc/logrotate.d/user_mgmt
/var/log/user_mgmt.log {
    daily
    rotate 30
    compress
    delaycompress
    missingok
    notifempty
    create 644 root root
    postrotate
        systemctl reload rsyslog >/dev/null 2>&1 || true
    endscript
}
```

Manual Log Cleanup

```
# Compress old logs
sudo gzip /var/log/user_mgmt.log.1

# Delete logs older than 90 days
sudo find /var/log/security_reports -name "*.txt" -mtime +90 -delete

# Archive before deletion
sudo tar -czf /archive/logs_$(date +%Y%m).tar.gz /var/log/security_reports/*.txt
```

10. Appendices

Appendix A: Command Quick Reference

User Management Commands

```
# Create user
create_user username [password]

# Delete user
delete_user [username]

# Update password
update_user [username] [new_password]

# Lock/unlock
lock_user [username]
unlock_user username
```

Analysis Commands

```
# Analyze logs
analyze_failed_logins

# Detect anomalies
detect_security_anomalies

# Generate report
generate_daily_report

# View logs
view_logs
```

Configuration Commands

```
# Setup FTP
setup_secure_ftp_server

# Enforce policies
set_password_policies

# Start monitoring
auto_monitor
```

Appendix B: Regular Expressions Used

Username Validation

```
^[a-z_][a-z0-9_-]*$
```

Password Validation

```
[A-Z]          # Uppercase
[a-z]          # Lowercase
[0-9]          # Number
[!@#$%^&*()_+] # Special character
```

Log Parsing

```
Failed password # Match failed attempts
Invalid user    # Match invalid user attempts
sudo.*FAILED    # Match sudo failures
```

Appendix C: Exit Codes Reference

Code	Meaning	Recovery Action
0	Success	Continue normal operation
1	General error	Check error message
2	Invalid input	Re-enter valid input
3	Permission denied	Run with sudo
4	File not found	Create required file
5	User exists	Choose different username
6	Password invalid	Choose stronger password

Appendix D: Environment Variables

Variable	Default	Purpose
PATH	/usr/local/bin:/usr/bin:/bin	Command search path
HOME	/home/username	User home directory
SHELL	/bin/bash	Default shell
LANG	en_US.UTF-8	Language setting

Appendix E: Testing Checklist

Pre-deployment Testing

- ☐ Verify script syntax: `bash -n security_system.sh`
- ☐ Check dependencies: `command -v useradd vsftpd`
- ☐ Test in VM first
- ☐ Create backup of existing configuration
- ☐ Verify log file permissions
- ☐ Test FTP connectivity
- ☐ Validate password policy

Production Testing

- ☐ Create test user
- ☐ Verify login logging
- ☐ Test report generation
- ☐ Verify email alerts
- ☐ Test backup restoration
- ☐ Validate log rotation
- ☐ Test auto-lock feature

Appendix F: Glossary

Term	Definition
Brute Force Attack	Repeated login attempts to guess passwords
Chroot Jail	Restricts FTP users to their home directory
Daemon	Background service process
Failed Login	Unsuccessful authentication attempt
Log Rotation	Automatic archiving of old log files
PAM	Pluggable Authentication Modules
SELinux	Security-Enhanced Linux
TLS/SSL	Transport Layer Security/Secure Sockets Layer
vsftpd	Very Secure FTP Daemon

Appendix G: Support Matrix

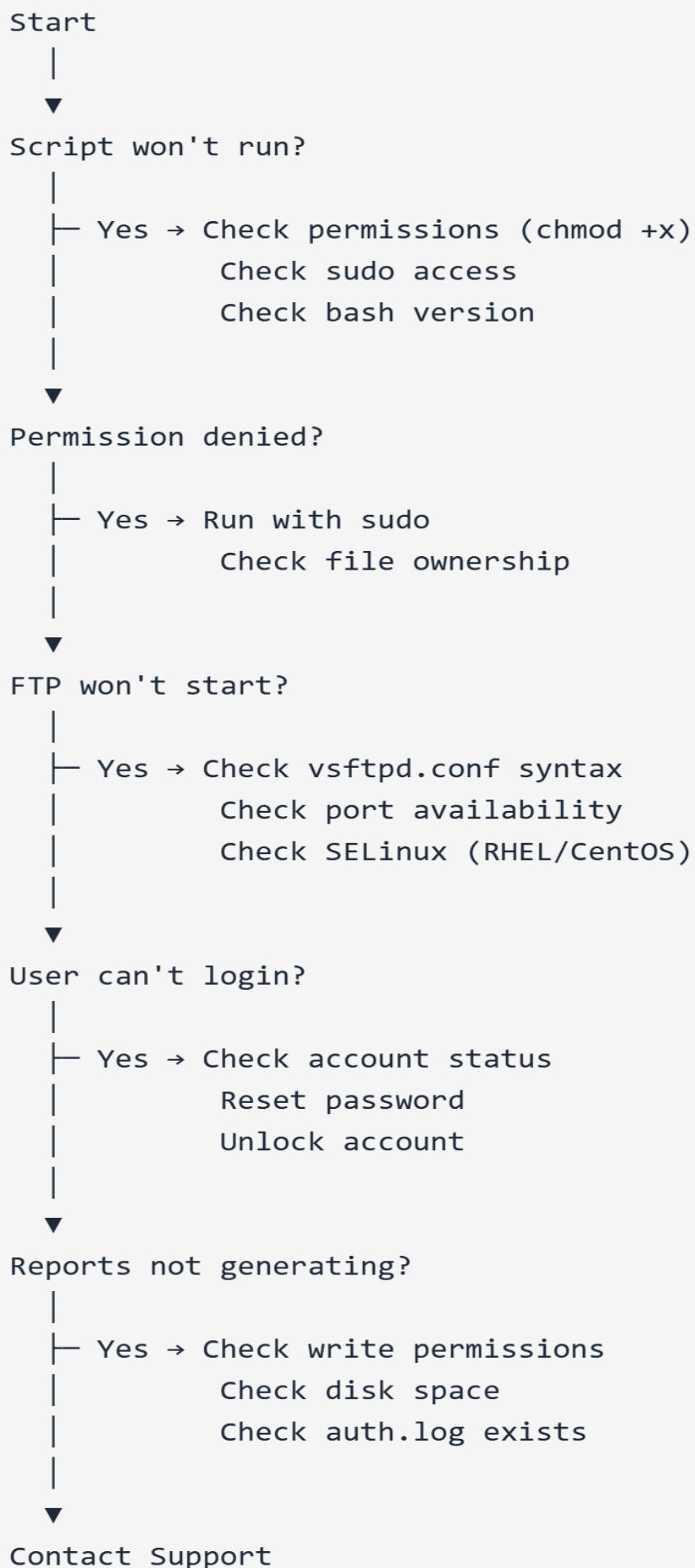
Operating System Support

OS	Version	Support Level	Notes
Ubuntu	18.04 LTS	Full	Tested
Ubuntu	20.04 LTS	Full	Recommended
Ubuntu	22.04 LTS	Full	Tested
Debian	10	Full	Tested
Debian	11	Full	Tested
CentOS	7	Full	Requires EPEL
CentOS	8	Full	Tested
RHEL	7	Full	Requires EPEL
RHEL	8	Full	Tested

Browser Support (HTML Reports)

Browser	Version	Support
Chrome	90+	Full
Firefox	88+	Full
Safari	14+	Full
Edge	90+	Full

Appendix H: Troubleshooting Flowchart



Appendix I: Sample Outputs

Sample Failed Login Report

```
=== Failed Login Attempts Report ===
Date: Mon Jan 15 10:30:45 EST 2024
=====

Failed Logins by IP Address:
    45 192.168.1.100
    23 10.0.0.50
    12 172.16.1.200

=== Suspicious IP Addresses (5+ failures) ===
192.168.1.100
10.0.0.50

=== Brute Force Detection ===
Possible brute force attacks detected:
    15 Jan 15 10:20:00
    12 Jan 15 10:25:00

=== Daily Summary ===
Total failed attempts: 80
Unique attacking IPs: 15
```

Sample HTML Report Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>Daily Security Report</title>
  <style>
    .header { background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); }
    .critical { background-color: #fee; border-left: 4px solid #f00; }
    .warning { background-color: #ffe; border-left: 4px solid #ff0; }
  </style>
</head>
<body>
  <div class="header">
    <h1>Daily Security Report</h1>
    <p>Date: Mon Jan 15 2024</p>
  </div>
  <!-- Report content -->
</body>
</html>
```

Appendix J: License Information

MIT License

Copyright (c) 2024 Security System Authors

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files...

Full license text available in LICENSE file.

Document Control

Version	Date	Author	Changes
1.0	2024-01-15	Security Team	Initial release
1.1	2024-02-01	Security Team	Added troubleshooting section
1.2	2024-03-01	Security Team	Updated API documentation

Documentation End

For support, updates, and contributions, please visit:

- GitHub: <https://github.com/Mood5-SUT/Secure-User-Log-Management>
- Documentation: <https://github.com/Mood5-SUT/Secure-User-Log-Management/blob/main/Documentation.pdf>

This documentation is maintained by the Security System Development Team.
Last Updated: January 2024